

Objective function

Differentiability: the second order

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Optimization: principles and algorithms

Differentiability: the second order

*The first derivatives provide information about the slope of the objective function.
The second derivatives provide information about its curvature.*

Second derivatives

Definition

$$\nabla f(x) : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

The gradient matrix of $\nabla f(x)$ is called the Hessian or the second derivatives matrix when it exists:

$$\nabla(\nabla f(x)) = \nabla^2 f(x).$$

Entries of the Hessian

$$(\nabla^2 f(x))_{ij} = \frac{\partial \nabla_i f(x)}{\partial x_j} = \frac{\partial (\partial f(x) / \partial x_i)}{\partial x_j} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$$

Properties

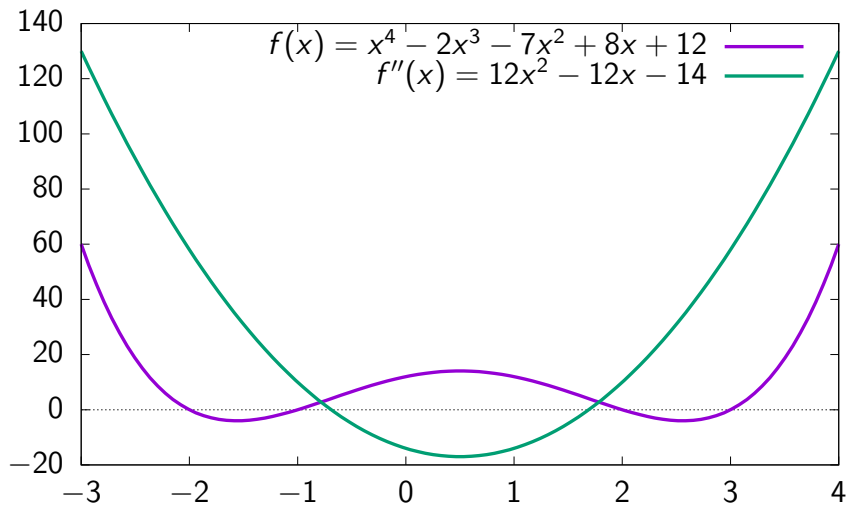
Symmetric

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$$

Convexity

If $\nabla^2 f(x)$ is positive semidefinite on $X \subseteq \mathbb{R}^n$, then f is convex on X .

Convexity



Geometry

One dimension

Follow direction d from x :

$$g(\alpha) = f(x + \alpha d)$$

$$g'(\alpha) = d^T \nabla f(x + \alpha d)$$

$$g''(\alpha) = d^T \nabla^2 f(x + \alpha d) d$$

Interpretation

$$g'(0) = d^T \nabla f(x)$$

directional derivative

$$g''(0) = d^T \nabla^2 f(x) d$$

directional curvature

Curvature

The curvature of f in x in direction d is defined as

$$\frac{d^T \nabla^2 f(x) d}{d^T d}.$$

Summary

The derivatives provide information on the geometry of the function

First: slope, second: curvature

The Hessian is symmetric

If positive semidefinite, then function is convex.